Table 1: Supported datasets (pre-build, 2026-07-14) schematized + L0–L4 verified in the TorchCell database. *Instances* is the dataset length (total genotype×environment records); *Shape* the shape of one phenotype instance; *Graph role* where the label sits in the cell graph; *Signal (gzip, bytes)* a Kolmogorov-complexity proxy – the gzip size in bytes of the concatenated per-record stored instances (perturbation + environment + phenotype), so a natural isolate's genome-defining perturbation set counts alongside its phenotype; the shared reference genome is never counted.

Dataset	Genotypes	Env	Instances	Phenotype	Shape	Graph role	Signal (gzip, bytes)
Fitness + genetic interaction							
Costanzo 2016 smf	20,484	2	20,484	single-mutant fitness	scalar	global	4.9×10^5
Costanzo 2016 dmf	20.7M	2	20,705,612	double-mutant fitness	scalar	global	6.5×10^8
Costanzo 2016 dmi	20.7M	2	20,705,612	digenic interaction	scalar	edge	6.7×10^8
Kuzmin 2018 smf	1,539	1	1,539	single-mutant fitness	scalar	global	3.9×10^4
Kuzmin 2018 dmf	410,399	1	410,399	double-mutant fitness	scalar	global	1.3×10^7
Kuzmin 2018 tmf	91,111	1	91,111	triple-mutant fitness	scalar	global	3.4×10^6
Kuzmin 2018 dmi	410,399	1	410,399	digenic interaction	scalar	edge	1.4×10^7
Kuzmin 2018 tmi	91,111	1	91,111	trigenic interaction	scalar	hyperedge	3.6×10^6
Kuzmin 2020 smf	472	1	472	single-mutant fitness	scalar	global	9.2×10^3
Kuzmin 2020 dmf	632,797	1	632,797	double-mutant fitness	scalar	global	2.0×10^7
Kuzmin 2020 tmf	301,798	1	301,798	triple-mutant fitness	scalar	global	1.1×10^7
Kuzmin 2020 dmi	632,797	1	632,797	digenic interaction	scalar	edge	2.2×10^7
Kuzmin 2020 tmi	301,798	1	301,798	trigenic interaction	scalar	hyperedge	1.2×10^7
Baryshnikova 2010 (smf)	5,993	1	5,993	single-mutant fitness	scalar	global	2.1×10^5
O'Duibhir 2014 (smf)	1,312	1	1,312	single-mutant fitness	scalar	global	3.3×10^4
Environmental / chemogenomic							
Auesukaree 2009 (stress screen)	333	6	525	stress sensitivity (categorical)	scalar	global	1.1×10^4
Mota 2024 (weak-acid screen)	601	3	1,273	weak-acid susceptibility (categorical)	scalar	global	2.4×10^4
Vanacloig-Pedros 2022	3,647	45	164,115	chemogenomic fitness (log2-ratio)	scalar	global	1.0×10^7
Costanzo 2021 (condition-SGA)	4,399	14	61,318	differential mutant fitness	scalar	global	1.1×10^6
Hillenmeyer 2008 het (FitDb HIP)	5,814	514	2,921,078	HIP fitness-defect log2-ratio	scalar	global	1.0×10^8
Hillenmeyer 2008 hom (FitDb HOP)	4,667	279	1,179,520	HOP fitness-defect z-score	scalar	global	4.3×10^7
Wildenhain 2015 (drug tolerance)	256	5,178	428,573	growth-inhibition z-score	scalar	global	1.9×10^7
Hoepfner 2014 (HIP/HOP atlas)	10,719	5,879	29,996,238	HIP/HOP sensitivity score	scalar	global	1.1×10^9
Smith 2006 (chemogenomic)	4,721	3	14,163	chemogenomic sensitivity (clear-zone ordinal)	scalar	global	2.1×10^5
Lian 2019 (MAGIC CRISPR-AID)	266,415	3	266,415	furfural tolerance fitness (log2-ratio)	scalar	global	1.6×10^7
Mormino 2022 (CRISPRi acetic-acid)	12	1	12	acetic-acid sensitivity (categorical)	scalar	global	1.1×10^3
Smith 2016 (CRISPRi chem-genetic)	1,035	26	14,463	chemogenomic fitness (log2-ratio)	scalar	global	6.0×10^5
Viability							
SGD essentiality	1,329	1	1,329	gene essentiality	scalar	node	1.5×10^4
SynLethDB (lethal)	14,000	1	14,000	synthetic lethality	scalar	edge	2.7×10^5
SynLethDB (rescue)	6,948	1	6,948	synthetic rescue	scalar	edge	1.4×10^5
Morphology							
Ohya 2005 (SCMD CalMorph)	4,718	1	4,718	cell morphology (CalMorph)	vector (281)	global	1.9×10^7
Ohnuki 2018 (SCMD CalMorph)	1,112	1	1,112	cell morphology (CalMorph)	vector (281)	global	5.9×10^6
Ohnuki 2022 (SCMD CalMorph)	1,979	1	1,979	cell morphology (CalMorph)	vector (281)	global	1.1×10^7
Expression (microarray)							
Kemmeren 2014	1,484	1	1,484	mRNA log2(mut/wt)	vector (6169)	node	4.0×10^8
Sameith 2015 sm	82	1	82	mRNA log2(mut/ref)	vector (6169)	node	2.3×10^7
Sameith 2015 dm	72	1	72	mRNA log2(mut/ref)	vector (6169)	node	2.1×10^7
Expression (RNA-seq)							
Caudal 2024 (pan-transcriptome)	943	1	943	mRNA abundance (RNA-seq)	vector (6000)	node	1.9×10^8
Nadal-Ribelles 2025 (Perturb-seq)	3,150	2	6,188	mRNA logFC (Perturb-seq)	vector (5639)	node	2.8×10^8
Metabolite							
Cachera 2023 (CRI-SPA betaxanthin)	4,735	1	4,735	betaxanthin (product proxy)	scalar	bipartite node	2.7×10^5
Mülleder 2016 (amino-acid metabolome)	4,678	1	4,678	amino-acid concentrations	vector (19)	bipartite node	9.9×10^5
Zelezniak 2018 (metabolome)	95	1	95	metabolite levels	vector (25)	bipartite node	3.8×10^4
Ozaydin 2013 (β-carotene screen)	4,474	1	4,474	β-carotene (colony-color visual score)	scalar	global	1.2×10^5
da Silveira 2014 (lipidomics)	127	1	127	lipid-species relative abundance	vector (135)	bipartite node	1.2×10^5
Yoshida 2012 (organic acids)	17	1	17	organic-acid titer	vector (6)	bipartite node	1.7×10^3
Xue 2025 (free fatty acids, private)	176	1	176	free-fatty-acid titer	vector (5)	bipartite node	2.2×10^4
Lopez 2024 (isobutanol screen, private)	4,554	1	4,554	isobutanol biosensor fold-change	scalar	bipartite node	1.1×10^5
Lopez 2024 (isobutanol validated, private)	224	1	224	isobutanol biosensor fold-change (validated)	scalar	bipartite node	9.3×10^3
Protein abundance							
Zelezniak 2018 (SWATH proteome)	97	1	97	protein abundance	vector (726)	node	2.1×10^6
Messner 2023 (proteome)	4,699	1	4,699	protein abundance	vector (1830)	node	1.4×10^8
Total (49 datasets)			**79,423,658**				3.8×10^9

In scope but not yet built/verified (no signal): Ho 2009 (bioethanol tolerance – genotype needs a WGS pipeline); Trikka 2015 (sclareol titers – figure-only data).